VERIDIAN PHARMA LTD. — Quality & Compliance Directorate, Hyderabad Facility

Standard Operating Procedure: Out of Specification (OOS) Investigation

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Owner: Arjun Nair (Lab Manager)	**Department:** Laboratory Operations	**Status:** Effective

1. Purpose

This standard operating procedure delineates the systematic, scientifically sound, and cGMP-compliant methodology for investigating, evaluating, and documenting Out of Specification (OOS) and atypical analytical test results obtained during the analysis of raw materials, in-process materials, finished products, and stability samples at Veridian Pharma Ltd., Hyderabad.

The objective is to establish whether an out-of-specification result is attributable to a laboratory-based execution error (analytical error, calculation error, standard degradation, equipment malfunction) or reflects a genuine manufacturing defect.

This procedure strictly follows US FDA Guidance for Industry: Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production and MHRA guidelines.

2. Scope

Applies to all analytical chemistry, microbiology, and physical testing laboratories supporting cGMP commercial production and stability programs.

Does not apply to non-commercial early developmental candidate screening where specifications have not yet been registered.

All analytical chemists, microbiologists, laboratory supervisors, and quality assurance release officers are bound by this procedure.

3. Responsibilities

• **Testing Analyst:** Immediately stop testing upon encountering an unexpected or out-of-specification result, preserve test solutions, and inform the supervisor without discarding consumables.

• **Lab Manager (Arjun Nair):** Lead the Phase I laboratory investigation, review raw chromatographic data, interview the testing chemist, and ensure adherence to investigation timelines.

• **QA Head (Meera Rao):** Authorize progression into Phase II investigations, evaluate and approve retesting protocols, and make final batch disposition determinations.

4. Procedure

4.1 Preservation of Test Materials: When an analytical result falls outside registered specification limits, the analyst must preserve all original solutions, standards, and instrument setups without alteration.

4.2 Immediate Supervisory Notification: The analyst must inform the Lab Manager within the same shift and initiate an OOS investigation form to establish contemporaneous documentation.

4.3 Phase I Laboratory Investigation Window: Phase I lab investigation completed within 3 working days.

4.4 Phase I Technical Review Scope: The Phase I investigation focuses on determining whether an obvious laboratory error occurred (e.g., calculation, dilution, instrument malfunction) by examining glassware, reagents, system suitability, and pipette calibrations.

4.5 Phase II Manufacturing Investigation Window: Phase II (full investigation) within 15 working days.

4.6 Phase II Cross-Functional Scope: Phase II encompasses a thorough cross-functional review of manufacturing records, environmental conditions, and raw materials involving production, engineering, and quality assurance.

4.7 Strict Retesting Governance: Retesting only with QA approval.

4.8 Predetermined Retest Protocol: Retest plans must define the exact number of replicate determinations and predetermined acceptance criteria before any retesting commences.

4.9 Prohibition of Averaging Across Failing Results: Averaging of original failing results with retest passing results is strictly prohibited to prevent masking true product heterogeneity.

4.10 Final Dossier Compilation and Authorization: The final investigation report must establish root cause, assign corrective actions, and receive final QA Head approval prior to batch disposition.

5. Records and Documentation

OOS Investigation Dossier (Form LAB-OOS-01), primary chromatographic integration logs, spectra printouts, supervisor interview checklists, and retest raw data sheets must be retained with the batch release file.

6. Revision History

Version	Effective Date	Change Description
1.0	2026-06-02	Initial release of OOS investigation procedure.